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2-3 DONNAN EQUILIBRIUM

2-3-1 DEFINITIONS

↪ The faraday (F) : the electric charge carried by one mole of electrons.

$$1F = \mathcal{N}_A \times e = 96500C$$

↪ The equivalent (Eq) : the amount of a substance that carries one faraday of electric charge.

Examples:

$1mole$ of $Na^+ \rightarrow 1Eq$

$1mole$ of $Ca^{2+} \rightarrow 2Eq$

↪ C_e : equivalent concentration of an ion i with valence z_i and concentration C_i,

$C_e = |z_i|C_i \qquad [C_e] = Eq/l$

Example:

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In a solution, $[Al^{3+}] = 1mol/l$

$z_{Al^{3+}} = 3 \Rightarrow C_e(Al^{3+}) = 3Eq/l$

↪ The total equivalent concentration of a solution:

$$C_e = C_e^+ + C_e^-,$$

with

$C_e^- = \sum_i |z_i^-|C_i^-$: the total equivalent concentration of anions,

$C_e^+ = \sum_i |z_i^+|C_i^+$: the total equivalent concentration of cations.

↪ The electroneutrality condition in a solution is written as:

$$C_e^+ = C_e^-$$
$$\Rightarrow C_e = 2C_e^+ = 2C_e^-$$

Exercise 12:

$1l$ of a solution contains $2moles$ of $CaCl_2$ and $1mole$ of $NaOH$. Calculate the equivalent concentration of this solution.

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$\underline{CaCl_2}$

$[CaCl_2] = 2mol/l$

$CaCl_2 \rightarrow Ca^{2+} + 2Cl^- \Rightarrow [Ca^{2+}] = 2mol/l; [Cl^-] = 4mol/l$

$\underline{NaOH}$

$[NaOH] = 1mol/l$

$NaOH \rightarrow Na^+ + OH^- \Rightarrow [Na^+] = 1mol/l; [OH^-] = 1mol/l$

$C_e^+ = z_{Ca^{2+}}[Ca^{2+}] + z_{Na^+}[Na^+] = 2 \times 2 + 1 \times 1 = 5Eq/l$

$C_e^- = |z_{Cl^-}|[Cl^-] + |z_{OH^-}|[OH^-] = 1 \times 4 + 1 \times 1 = 5Eq/l$

$C_e = 2C_e^+ = 10Eq/l$

2-3-2 NERNST EQUATION

We consider two compartments (1) and (2) of equal volume V, separated by a non-sliding dialysis membrane.

In (1) : salt AB, with $[AB]_1 = C$.

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In (2) : salt AB and protein $PrBz$, with $[AB]_2 = C$, $[PrBz]_2 = C'$

Initially

↪ In (1) : $AB \rightarrow A^- + B^+$

A^- and B^+ diffusibles.

↪ In (2) : $AB \rightarrow A^- + B^+$,

$PrBz \rightarrow Pr^{z-} + zB^+$

Pr^{z-} non diffusible.

Initially	
(1)	(2)
$[A^-]_1 = C$ $[B^+]_1 = C$	$[A^-]_2 = C$ $[B^+]_2 = C + zC'$ $[Pr^{z-}]_2 = C'$

$[B^+]_2 > [B^+]_1 \Rightarrow n$ moles of B^+ at a concentration of $x = \frac{n}{V}$ diffuse from $(2) \rightarrow (1)$. They will be accompanied by n moles of A^- to ensure electroneutrality.

At equilibrium

At equilibrium, we notice that $[A^-]_1 > [A^-]_2$, and we verify that $[B^+]_2 > [B^+]_1$.

At equilibrium	
(1)	(2)
$[B^+]_1 = C + x$ $[A^-]_1 = C + x$	$[B^+]_2 = C + zC' - x$ $[A^-]_2 = C - x$ $[Pr^{z-}]_2 = C'$

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In fact, everything happens as if the macro-ions Pr^{z-} retained in (2) :

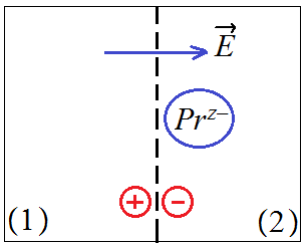
⇢ attract B^+ through a force $\overrightarrow{F_+} = q_{B^+}\vec{E}$

⇢ repel A^- through a force $\overrightarrow{F_-} = q_A\ \vec{E}$

$\vec{E}$: ***transmembranaire*** electric field.

The electric field always points from $(+) \rightarrow (-)$.

Therefore, the side of the membrane in contact with Pr^{z-} is negatively charged, while the other side is positively charged.



$\vec{E} : (1) \rightarrow (2) \Rightarrow V_1 > V_2$

We define,

$$E = -\frac{V_2 - V_1}{l}$$

$V_2 - V_1$: membrane potential.

V_2 and V_1: electrochemical potentials.

l : membrane thickness.

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It is shown that:

$$V_2 - V_1 = -\frac{RT}{z\mathcal{F}} \ln \frac{C_2}{C_1}$$: ***Nernst equation*** for diffusible ions.

Or equivallently,

$$V_2 - V_1 = -\frac{kT}{ze} \ln \frac{C_2}{C_1}$$

Since, $\mathcal{F} = \mathcal{N}_A \times e$ and $R = \mathcal{N}_A \times k$, thus, $\frac{kT}{ze} = \frac{RT}{z\mathcal{F}}$

2-3-3 DONNAN EQUILIBRIUM EQUATION

Ion I of valence $z \rightsquigarrow V_2 - V_1 = -\frac{kT}{ze} \ln \frac{[I]_2}{[I]_1}$

Ion I' of valence $z' \rightsquigarrow V_2 - V_1 = -\frac{kT}{z'e} \ln \frac{[I']_2}{[I']_1}$

Therefore,

$$\frac{1}{z} \ln \frac{[I]_2}{[I]_1} = \frac{1}{z'} \ln \frac{[I']_2}{[I']_1}$$: ***Donnan equilibrium*** equation or ***Gibbs-Donnan*** equation

Or equivallently,

$$\frac{1}{z}\ln\frac{C_{e_2}(I)}{C_{e_1}(I)} = \frac{1}{z'}\ln\frac{C_{e_2}(I')}{C_{e_1}(I')}$$

Since, $C_e(I) = |z|[I] \Rightarrow \frac{[I]_2}{[I]_1} = \frac{C_{e_2}(I)}{C_{e_1}(I)}$

Application to ions A^- $(z_{A^-} = -1)$ *and* $B^+(z_{B^+} = +1)$:

$$\frac{1}{z}\ln\frac{[I]_2}{[I]_1} = \frac{1}{z'}\ln\frac{[I']_2}{[I']_1}$$

$$\Rightarrow \frac{1}{z_{A^-}}\ln\frac{[A^-]_2}{[A^-]_1} = \frac{1}{z_{B^+}}\ln\frac{[B^+]_2}{[B^+]_1}$$

$$\Rightarrow \ln\frac{[A^-]_2}{[A^-]_1} = -\ln\frac{[B^+]_2}{[B^+]_1} = \ln\frac{[B^+]_1}{[B^+]_2}$$

$$\Rightarrow \frac{[A^-]_2}{[A^-]_1} = \frac{[B^+]_1}{[B^+]_2}$$

$$[A^-]_1 > [A^-]_2 \Rightarrow [B^+]_2 > [B^+]_1$$

$$\frac{[A^-]_2}{[A^-]_1} = \frac{[B^+]_1}{[B^+]_2} \Rightarrow \frac{C-x}{C+x} = \frac{C+x}{C-x+zC'} \Rightarrow x = \frac{zCC'}{4C+zC'}$$

At equilibrium	
$[A^-]_1 = C + x$	$[A^-]_2 = C - x$
$[B^+]_1 = C + x$	$[B^+]_2 = C + zC' - x$
	$[Pr^{z-}]_2 = C'$